

Shunya Voice Agents

Partner Integration Guide

Use this document to integrate with the Shunya voice agent platform without using the dashboard.

Item	Value
Base URL	<code>https://agents.shunyalabs.ai</code>
Note	Shunya will provide your API key before go-live. Use it on every request below.

Bring your own Plivo: If you bring your own Plivo account, register your caller ID via §7 Phone numbers before triggering calls. If Shunya provisions telephony for you, skip §7 and use the caller ID they provide.

Shunya-managed AI: For pipeline_type: "shunya", Shunya handles speech recognition, language model, and text-to-speech. Configure agent behaviour the same way you would on the dashboard — prompt, voice, messages, call behaviour, recording, outcomes, and webhooks. Do NOT send providers or tts_voice_id; those are internal and not shown on the dashboard.

Authentication

All API requests (except health check) require:

```
Authorization: Bearer YOUR_API_KEY
Content-Type: application/json
```

Replace `YOUR_API_KEY` with the key Shunya gives you. Keep it secret.

Endpoints

Method	Path	Description
GET	<code>/api/health</code>	Health check
POST	<code>/api/v1/agents</code>	Create agent
GET	<code>/api/v1/agents</code>	List agents
GET	<code>/api/v1/agents/{agent_id}</code>	Get one agent
PATCH	<code>/api/v1/agents/{agent_id}</code>	Update agent

Method	Path	Description
DELETE	/api/v1/agents/{agent_id}	Archive agent
GET	/api/telephony/accounts	List telephony accounts
POST	/api/telephony/accounts	Create Plivo/Twilio account
GET	/api/telephony/accounts/{account_id}	Get one telephony account
PATCH	/api/telephony/accounts/{account_id}	Update account label / status
DELETE	/api/telephony/accounts/{account_id}	Disable telephony account
GET	/api/telephony/numbers	List registered phone numbers
POST	/api/telephony/numbers	Register phone number (new or existing account)
GET	/api/telephony/numbers/{number_id}	Get one phone number
PATCH	/api/telephony/numbers/{number_id}	Update number label / status
DELETE	/api/telephony/numbers/{number_id}	Remove phone number
POST	/api/v1/calls	Trigger outbound call

Quick reference — all curls

#	Action	Section
1	Health check	§1
2	Create agent (minimum)	§2
3	Create agent (full dashboard config)	§2
4	List agents	§3
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6	Update agent	§5
7	Update webhooks	§5
8	Clear webhooks	§5
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10	Register Plivo + number (one shot)	§7
11	List phone numbers	§7
12	Add number to existing account	§7
13	Trigger call (minimum)	§8
14	Trigger call (all fields)	§8

Note: Phone number endpoints live under `/api/telephony/*` (not `/api/v1/*`) but use the same API key as agent CRUD and calls.

Not on this API (contact Shunya): viewing call history in UI, billing, campaigns.

Call results are delivered via **webhooks** (agent-level `webhooks[]` or per-call `callback_url`) — there is no GET endpoint to poll call status.

1. Health check

```
curl -s 'https://agents.shunyalabs.ai/api/health'
```

Response: `{"status":"ok"}`

2. Create agent

Required fields: `name`, `prompt.system_prompt`, `messages.opening`

Minimum example

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/v1/agents' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "name": "My Voice Agent",
  "prompt": {
    "system_prompt": "You are a helpful assistant."
  },
  "messages": {
    "opening": "Hello, how can I help you today?"
  }
}'
```

Response (201): JSON with `"id"` — save this as `YOUR_AGENT_ID` for later calls.

Full dashboard config example

Omit any optional field you do not need — the API applies defaults. This example covers every agent setting available on the dashboard (Prompt, Behaviour, Recording, Outcomes, Advanced / pre-call API, Webhooks).

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/v1/agents' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "name": "Sales Agent",
  "description": "Outbound sales agent with full configuration",
  "status": "active",
  "pipeline_type": "shunya",
```

```
"prompt": {
  "system_prompt": "You are a polite sales assistant. Keep responses concise for phone calls.",
  "temperature": 0.7,
  "max_tokens": 1024,
  "extraction_schema": {
    "type": "object",
    "properties": {
      "customer_name": { "type": "string" },
      "interested_model": { "type": "string" },
      "call_outcome": { "type": "string" }
    }
  }
},
"voice": {
  "language": "hi",
  "tone": "professional",
  "tone_description": "Warm, confident sales tone",
  "accent": "indian",
  "accent_description": "Standard Indian Hindi accent",
  "gender": "female",
  "tts_speed": 1.0
},
"messages": {
  "opening": "Namaste! Main aapki kaise madad kar sakti hoon?",
  "closing": "Dhanyavaad, aapka din shubh ho!"
},
"behavior": {
  "supports_barge_in": true,
  "interruption_sensitivity": 0.5,
  "block_interruptions": false,
  "silence_timeout_secs": 10,
  "max_call_duration_secs": 600,
  "max_session_minutes": 30,
  "fast_farewell_enabled": false,
  "voicemail_detection_enabled": true,
  "voicemail_detection_delay_secs": 4,
  "voicemail_message": "Namaste, yeh ek important call hai. Kripya humein wapas call karein.",
  "idle_reprompt_message": "Kya aap ab bhi line par hain?",
  "idle_max_retries": 2,
  "idle_terminate_message": "Lagta hai aap busy hain. Hum baad mein call karenge. Dhanyavaad!",
  "pre_call_api_enabled": true,
  "pre_call_api_url": "https://your-server.example.com/api/pre-call",
  "pre_call_api_method": "POST",
  "pre_call_api_timeout_ms": 2000,
  "pre_call_api_headers": {
    "Authorization": "Bearer your-pre-call-token"
  },
  "call_direction": "outbound",
  "transfer_target": "+918031137171"
},
"recording": {
  "enabled": true,
  "enable_transcript": true,
  "enable_summary": true
},
"outcomes": [
  {
    "value": "interested",
    "description": "Customer showed interest in the product"
```

```

    },
    {
      "value": "callback_requested",
      "description": "Customer asked for a callback later"
    },
    {
      "value": "not_interested",
      "description": "Customer declined"
    },
    {
      "value": "wrong_number",
      "description": "Reached someone other than the intended customer"
    },
    {
      "value": "no_answer",
      "description": "Customer did not answer"
    }
  ],
  "webhooks": [
    {
      "event_type": "call.completed",
      "url": "https://your-server.example.com/webhooks/call-completed",
      "secret": "your-webhook-secret16",
      "description": "Call completion webhook"
    }
  ]
}'

```

Field reference — top level

Field	Required	Description
name	Yes	Agent display name (1–255 chars)
description	No	Optional description
status	No	draft, active, archived (default active)
pipeline_type	No	"shunya" only (default)
prompt	Yes	System prompt config (see below)
voice	No	Voice / language settings
messages	Yes	Opening (and optional closing) messages
behavior	No	Call behavior, barge-in, silence, voicemail, pre-call API
recording	No	Recording, transcript, summary toggles
outcomes	No	Custom call outcome labels (max 20)
webhooks	No	Agent-level webhook subscriptions

Field reference — prompt

Field	Type	Range / notes
system_prompt	string	Required. Agent instructions
temperature	float	0.0–2.0
max_tokens	int	1–8192
extraction_schema	object	JSON schema for post-call data extraction

Field reference — voice

Field	Type	Values / range
language	string	en, hi, hinglish, ta, te, bn, mr, gu
tone	string	neutral, warm, professional, casual, assertive
tone_description	string	Free text (max 8000 chars)
accent	string	neutral, indian, british, american, australian
accent_description	string	Free text (max 1000 chars)
gender	string	female, male, neutral
tts_speed	float	0.5–2.0 (default 1.0)

Field reference — messages

Field	Type	Notes
opening	string	Required. First message the agent speaks
closing	string or object	Goodbye message

Field reference — behavior

Field	Type	Range	Description
supports_barge_in	bool	—	Allow caller to interrupt the agent
interruption_sensitivity	float	0.0–1.0	How sensitive barge-in is
block_interruptions	bool	—	Block all caller interruptions
silence_timeout_secs	int	1–120	Seconds of silence before reprompt
max_call_duration_secs	int	30–7200	Hard cap on call length (seconds)
max_session_minutes	int	1–120	Max session duration (minutes)
fast_farewell_enabled	bool	—	Short goodbye when call ends quickly
voicemail_detection_enabled	bool	—	Detect and handle voicemail
voicemail_detection_delay_secs	int	1–30	Delay before voicemail detection

Field	Type	Range	Description
voicemail_message	string	max 500	Message left on voicemail
idle_reprompt_message	string	max 500	Reprompt when caller is silent
idle_max_retries	int	0–10	Max silence reprompt retries
idle_terminate_message	string	max 500	Message before hanging up on idle
pre_call_api_enabled	bool	—	Call external API before dialing
pre_call_api_url	string	max 2000	Pre-call API endpoint URL
pre_call_api_method	string	GET, POST	HTTP method for pre-call API
pre_call_api_timeout_ms	int	100–10000	Pre-call API timeout (ms)
pre_call_api_headers	object	—	Headers sent to pre-call API
call_direction	string	outbound	Call direction
transfer_target	string	max 200	Number or target for transfers

Field reference — recording

Field	Type	Description
enabled	bool	Record calls
enable_transcript	bool	Generate transcript
enable_summary	bool	Generate call summary

Field reference — outcomes[]

Field	Type	Description
value	string	Outcome label (1–64 chars)
description	string	Plain-English meaning for the extractor (max 500)

Field reference — webhooks[]

Field	Type	Description
event_type	string	Default call.completed
url	string (URL)	Webhook destination
secret	string	HMAC secret (min 16 chars)
description	string	Optional label

3. List agents

```
curl -s 'https://agents.shunyalabs.ai/api/v1/agents' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

With filters:

```
curl -s 'https://agents.shunyalabs.ai/api/v1/agents?status=active&limit=50&offset=0' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

Response (200):

```
{
  "agents": [ { "id": "...", "name": "...", "status": "active", ... } ],
  "total": 1
}
```

4. Get one agent

```
curl -s 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

5. Update agent

Only include fields you want to change.

```
curl -s -X PATCH 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "name": "Sales Agent v2",
  "messages": {
    "closing": "Thank you, goodbye!"
  }
}'
```

Update webhooks (replaces the full webhook list):

```
curl -s -X PATCH 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "webhooks": [
    {
      "event_type": "call.completed",
```



```

    "url": "https://your-server.example.com/webhooks/updated",
    "secret": "new-secret-min16ch",
    "description": "Updated webhook"
  }
]
}'

```

Clear all webhooks:

```

curl -s -X PATCH 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{ "webhooks": [] }'

```

Full dashboard config PATCH (only supplied fields are updated):

```

curl -s -X PATCH 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "name": "Sales Agent v2",
  "description": "Updated description",
  "status": "active",
  "pipeline_type": "shunya",
  "prompt": {
    "system_prompt": "Updated system prompt.",
    "temperature": 0.6,
    "max_tokens": 800,
    "extraction_schema": {
      "type": "object",
      "properties": {
        "call_outcome": { "type": "string" }
      }
    }
  },
  "voice": {
    "language": "hi",
    "tone": "warm",
    "tone_description": "Friendly tone",
    "accent": "indian",
    "accent_description": "Indian Hindi",
    "gender": "female",
    "tts_speed": 1.05
  },
  "messages": {
    "opening": "Namaste! Updated opening message.",
    "closing": "Dhanyavaad, phir milenge!"
  },
  "behavior": {
    "supports_barge_in": true,
    "interruption_sensitivity": 0.6,
    "block_interruptions": false,
    "silence_timeout_secs": 12,
    "max_call_duration_secs": 900,
    "max_session_minutes": 45,
    "fast_farewell_enabled": true,
    "voicemail_detection_enabled": true,

```

```

    "voicemail_detection_delay_secs": 5,
    "voicemail_message": "Please call us back.",
    "idle_reprompt_message": "Are you still there?",
    "idle_max_retries": 3,
    "idle_terminate_message": "We will call back later.",
    "pre_call_api_enabled": false,
    "pre_call_api_url": "https://your-server.example.com/api/pre-call",
    "pre_call_api_method": "POST",
    "pre_call_api_timeout_ms": 3000,
    "pre_call_api_headers": {},
    "call_direction": "outbound",
    "transfer_target": "+918031137171"
  },
  "recording": {
    "enabled": true,
    "enable_transcript": true,
    "enable_summary": true
  },
  "outcomes": [
    { "value": "interested", "description": "Customer interested" }
  ],
  "webhooks": []
}'

```

6. Delete agent

Soft-deletes the agent (status becomes `archived`).

```

curl -s -X DELETE 'https://agents.shunyalabs.ai/api/v1/agents/YOUR_AGENT_ID' \
-H 'Authorization: Bearer YOUR_API_KEY'

```

Response: 204 No Content

7. Phone numbers (telephony)

Register your Plivo caller ID before triggering outbound calls. This mirrors the dashboard Phone numbers page: connect your Plivo account, register E.164 numbers, list them, update labels, and remove them.

Uses the **same** `Authorization: Bearer YOUR_API_KEY` as `/api/v1/*`.

When you trigger a call, `from_number` must match an **active** number registered here (or omit `from_number` if your org has exactly one active number).

Credentials (`auth_id`, `auth_token`) are validated against Plivo at registration time, encrypted at rest, and never returned in API responses (only masked values).

Shunya-managed telephony: If Shunya provisions a shared caller ID for you (dashboard “Use our number”), you do not need to call these endpoints — go straight to §8 Trigger outbound call.

Register Plivo account + number (one shot)

Use when setting up for the first time — same as Add number → New Plivo account on the dashboard.

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/telephony/numbers' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "tag": "new",
  "provider": "plivo",
  "auth_id": "YOUR_PLIVO_AUTH_ID",
  "auth_token": "YOUR_PLIVO_AUTH_TOKEN",
  "phone_number": "+918031137171",
  "label": "Outbound caller ID",
  "account_label": "Our Plivo account"
}'
```

Response (201): phone number record with "id", "phone_number", "telephony_account_id", "is_active": true.

Two-step setup (account first, then number)

Step 1 — Create telephony account:

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/telephony/accounts' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "provider": "plivo",
  "auth_id": "YOUR_PLIVO_AUTH_ID",
  "auth_token": "YOUR_PLIVO_AUTH_TOKEN",
  "label": "Our Plivo account"
}'
```

Save "id" from the response as TELEPHONY_ACCOUNT_ID.

Step 2 — Add number to that account (same as Add number → Use existing account):

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/telephony/numbers' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "tag": "existing",
  "telephony_account_id": "TELEPHONY_ACCOUNT_ID",
  "phone_number": "+918031137171",
  "label": "Outbound caller ID"
}'
```

List phone numbers

```
curl -s 'https://agents.shunyalabs.ai/api/telephony/numbers' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

Include inactive numbers:

```
curl -s 'https://agents.shunyalabs.ai/api/telephony/numbers?include_inactive=true' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

Response (200):

```
{
  "numbers": [
    {
      "id": "...",
      "phone_number": "+918031137171",
      "provider": "plivo",
      "label": "Outbound caller ID",
      "is_active": true,
      "telephony_account_id": "..."
    }
  ],
  "total": 1
}
```

Get one phone number

```
curl -s 'https://agents.shunyalabs.ai/api/telephony/numbers/NUMBER_ID' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

Update phone number (label or active status)

```
curl -s -X PATCH 'https://agents.shunyalabs.ai/api/telephony/numbers/NUMBER_ID' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "label": "Updated label",
  "is_active": true
}'
```

Remove phone number

```
curl -s -X DELETE 'https://agents.shunyalabs.ai/api/telephony/numbers/NUMBER_ID' \
-H 'Authorization: Bearer YOUR_API_KEY'
```

Response: 204 No Content

List telephony accounts

```
curl -s 'https://agents.shunyalabs.ai/api/telephony/accounts' \
```

```
-H 'Authorization: Bearer YOUR_API_KEY'
```

Response (200): accounts with masked `auth_id_masked` / `auth_token_masked` (never raw credentials).

Get one telephony account

```
curl -s 'https://agents.shunyalabs.ai/api/telephony/accounts/TELEPHONY_ACCOUNT_ID' \  
-H 'Authorization: Bearer YOUR_API_KEY'
```

Update telephony account

```
curl -s -X PATCH 'https://agents.shunyalabs.ai/api/telephony/accounts/TELEPHONY_ACCOUNT_ID' \  
-H 'Authorization: Bearer YOUR_API_KEY' \  
-H 'Content-Type: application/json' \  
-d '{  
  "label": "Updated account label",  
  "is_active": true  
'
```

Disable telephony account

Removes the account when it has no active numbers (same as Disable on the dashboard).

```
curl -s -X DELETE 'https://agents.shunyalabs.ai/api/telephony/accounts/TELEPHONY_ACCOUNT_ID' \  
-H 'Authorization: Bearer YOUR_API_KEY'
```

Response: `204 No Content` — returns `409` if the account still has phone numbers attached.

Telephony field reference

POST /api/telephony/numbers — tag: "new"

Field	Required	Description
tag	Yes	Must be "new"
provider	Yes	"plivo" or "twilio"
auth_id	Yes	Plivo Auth ID
auth_token	Yes	Plivo Auth Token
phone_number	Yes	E.164 format, e.g. +918031137171
label	No	Friendly name for the number
account_label	No	Friendly name for the new account

POST /api/telephony/numbers — tag: "existing"

Field	Required	Description
tag	Yes	Must be "existing"
telephony_account_id	Yes	UUID from list/create accounts
phone_number	Yes	E.164 format
label	No	Friendly name for the number

POST /api/telephony/accounts

Field	Required	Description
provider	Yes	"plivo" or "twilio"
auth_id	Yes	Provider Auth ID
auth_token	Yes	Provider Auth Token
label	No	Friendly account name

8. Trigger outbound call

Queues an outbound phone call. Response is immediate; dialing runs in the background.

Minimum example

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/v1/calls' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "agent_id": "YOUR_AGENT_ID",
  "phone": "+919876543210",
  "from_number": "+918031137171",
  "external_id": "call-001"
}'
```

Complete example (all supported fields)

```
curl -s -X POST 'https://agents.shunyalabs.ai/api/v1/calls' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-H 'Content-Type: application/json' \
-d '{
  "agent_id": "YOUR_AGENT_ID",
  "phone": "+919876543210",
  "from_number": "+918031137171",
  "callback_url": "https://your-server.example.com/webhooks/call-callback",
  "callback_secret": "callback-secret-min16ch",
  "context": {
```

```

    "customer_name": "Rahul Sharma",
    "interested_model": "Swift",
    "city": "Delhi"
  },
  "echo": {
    "campaign": "test-drive",
    "source": "partner-api"
  },
  "external_id": "call-002",
  "start": 900,
  "end": 2100,
  "timezone": "Asia/Kolkata",
  "retries": 1
}'

```

Call fields

Field	Required	Type	Description
agent_id	Yes	UUID	Agent to use for the call
phone	Yes	string	Customer number, E.164 (+91...)
from_number	No*	string	Outbound caller ID (E.164)
callback_url	No	string	One-shot webhook URL when call completes
callback_secret	No	string	Accepted (min 16 chars); callbacks currently unsigned
context	No	object	Variables merged into agent customer_variables
echo	No	object	Returned verbatim in the call callback
external_id	No	string	Idempotency key (max 255 chars)
start	No**	int	Window start, HHMM (e.g. 900 = 09:00)
end	No**	int	Window end, HHMM (e.g. 2100 = 21:00)
timezone	No	string	IANA timezone (default Asia/Kolkata)
retries	No	int	Retry count (0–10, capped by agent config)

*Required if your account has multiple active outbound numbers.

**start and end must both be set or both omitted.

Response (201):

```

{
  "call_id": "3c2fc901-0c47-4735-b35f-efe2e0720328",
  "status": "queued",
  "external_id": "call-001",
  "duplicate": false
}

```

Use a **new external_id** for each new call. Reusing the same id returns the original call with `"duplicate": true`.

Typical flow (no dashboard)

BYO Plivo (bring your own telephony):

1. POST /api/telephony/numbers → register Plivo + caller ID (once)
2. GET /api/telephony/numbers → confirm number is active (optional)
3. POST /api/v1/agents → create agent, save id
4. POST /api/v1/calls → trigger call (from_number = registered number)
5. (webhook) → receive call outcome on your server

Shunya-managed telephony (caller ID pre-provisioned by Shunya):

1. POST /api/v1/agents → create agent, save id
2. POST /api/v1/calls → trigger call with agent id + customer phone
3. (webhook) → receive call outcome on your server
4. GET /api/v1/agents/{id} → verify agent config (optional)
5. PATCH /api/v1/agents/{id} → update agent (optional)

HTTP status codes

Code	Meaning
200	Success (GET, PATCH)
201	Created (agent, call, phone number, account)
204	Deleted (agent archived, phone number, account)
401	Invalid or missing API key
404	Agent, phone number, or account not found
409	Duplicate external_id on call; duplicate phone/account; account still has numbers
422	Invalid request body
400	Invalid Plivo credentials or number not on account
403	Outside allowed calling window (start/end)

Contact Shunya support. Can you make this in the proper document format with proper formatting and better writing? I will give you a sample of what if you receive repeated 500 or 503 responses.

Support

For API keys, caller ID numbers, or integration help, contact your Shunya account team.